

PHYS 812: Assignment 1

Problems and Solutions

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1. Consider the double delta-function potential

$$V(x) = -\alpha[\delta(x+a) + \delta(x-a)],$$

where α and a are positive constants. For the case of even wave function, how many bound states does this potential possess? Find the allowed energies for $\alpha = \hbar^2/(2ma)$ (in units $\hbar^2/(ma^2)$, iterate only once). Sketch the wave functions.

Solution.

For a bound state, write

$$E = -\frac{\hbar^2 \kappa^2}{2m}, \quad \kappa > 0.$$

Away from the delta functions, the Schrödinger equation becomes

$$\psi''(x) = \kappa^2 \psi(x).$$

Since the potential and the desired wave function are even, a normalizable solution may be written as

$$\psi_e(x) = \begin{cases} C e^{\kappa(x+a)}, & x < -a, \\ D \cosh(\kappa x), & -a \leq x \leq a, \\ C e^{-\kappa(x-a)}, & x > a. \end{cases}$$

Continuity at $x = a$ requires

$$C = D \cosh(\kappa a).$$

Integrating the Schrödinger equation across the delta function at $x = a$ gives the derivative discontinuity

$$\psi'(a^+) - \psi'(a^-) = -\frac{2m\alpha}{\hbar^2} \psi(a).$$

For the wave function above,

$$\psi'(a^+) = -\kappa C, \quad \psi'(a^-) = \kappa D \sinh(\kappa a), \quad \psi(a) = C.$$

Therefore,

$$-\kappa C - \kappa D \sinh(\kappa a) = -\frac{2m\alpha}{\hbar^2} C.$$

Using $C = D \cosh(\kappa a)$, we obtain

$$\kappa[1 + \tanh(\kappa a)] = \frac{2m\alpha}{\hbar^2},$$

or equivalently,

$$\boxed{\kappa = \frac{m\alpha}{\hbar^2} (1 + e^{-2\kappa a})}.$$

Introduce the dimensionless quantities

$$y = \kappa a, \quad \lambda = \frac{m\alpha a}{\hbar^2}.$$

The transcendental equation becomes

$$y = \lambda (1 + e^{-2y}).$$

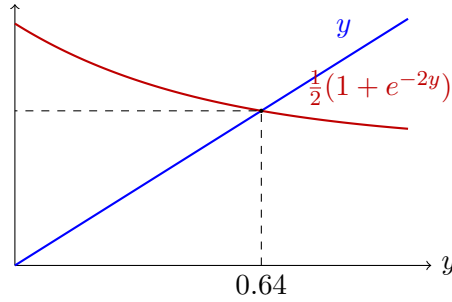
For

$$\alpha = \frac{\hbar^2}{2ma},$$

we have $\lambda = 1/2$, and hence

$$\boxed{y = \frac{1}{2} (1 + e^{-2y})}.$$

The left-hand side is strictly increasing from zero, while the right-hand side is strictly decreasing from 1 toward 1/2. Consequently, there is exactly one positive intersection and therefore exactly one even bound state.



The graph gives the starting approximation

$$y_0 \simeq 0.64.$$

Define

$$F(y) = y - \frac{1}{2} (1 + e^{-2y}), \quad F'(y) = 1 + e^{-2y}.$$

One Newton iteration gives

$$\begin{aligned} y_1 &= y_0 - \frac{F(y_0)}{F'(y_0)} \\ &= 0.64 - \frac{0.64 - \frac{1}{2}(1 + e^{-1.28})}{1 + e^{-1.28}} \\ &= 0.639232. \end{aligned}$$

Thus,

$$\kappa \simeq \frac{0.639232}{a}.$$

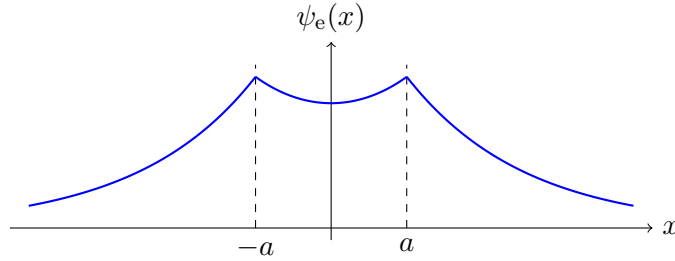
The energy is therefore

$$\begin{aligned} E &= -\frac{\hbar^2 \kappa^2}{2m} \\ &= -\frac{y_1^2}{2} \frac{\hbar^2}{ma^2} \\ &\simeq \boxed{-0.2043 \frac{\hbar^2}{ma^2}}. \end{aligned}$$

Up to normalization, the bound-state wave function is

$$\psi_e(x) = \mathcal{N} \begin{cases} \cosh(\kappa a) e^{\kappa(x+a)}, & x < -a, \\ \cosh(\kappa x), & |x| \leq a, \\ \cosh(\kappa a) e^{-\kappa(x-a)}, & x > a. \end{cases}$$

It is even and nodeless, has cusped maxima at $x = \pm a$, and decays exponentially outside the two delta functions.



For completeness, the odd-parity condition at this coupling would be

$$y = \frac{1}{2} (1 - e^{-2y}).$$

For every $y > 0$, the right-hand side is smaller than y , so there is no positive odd-state solution. The root $y = 0$ corresponds to $E = 0$ and is not normalizable. Thus, the potential has only the single even bound state.

2. Prove the (generalized) Heisenberg uncertainty relation for hermitean operators A and B :

$$(\Delta A)^2 (\Delta B)^2 \geq \frac{1}{4} |\langle [A, B] \rangle|^2,$$

where

$$\Delta C = \sqrt{\langle \psi | (C - \langle C \rangle)^2 | \psi \rangle}, \quad \langle C \rangle = \langle \psi | C | \psi \rangle.$$

- (a) Show that the commutator $[A, B]$ is antihermitean.
(b) Apply the Schwarz inequality,

$$\langle \chi_1 | \chi_1 \rangle \langle \chi_2 | \chi_2 \rangle \geq |\langle \chi_1 | \chi_2 \rangle|^2,$$

to the left-hand side of the uncertainty relation.

- (c) Use the identity

$$2AB = [A, B]_+ + [A, B],$$

where $[A, B]_+$ is the anticommutator, in the right-hand side of the Schwarz inequality.

- (d) Argue that the term containing the anticommutator is positive.

Solution.

Assume that $\langle \psi | \psi \rangle = 1$, and define the centered operators

$$\bar{A} = A - \langle A \rangle I, \quad \bar{B} = B - \langle B \rangle I,$$

where

$$\langle O \rangle = \langle \psi | O | \psi \rangle.$$

Since A and B are Hermitian, their expectation values are real. Consequently, $\bar{A}$ and $\bar{B}$ are also Hermitian. Moreover,

$$(\Delta A)^2 = \langle \bar{A}^2 \rangle, \quad (\Delta B)^2 = \langle \bar{B}^2 \rangle.$$

(a) The commutator is anti-Hermitian.

Using $A^\dagger = A$ and $B^\dagger = B$,

$$\begin{aligned} [A, B]^\dagger &= (AB - BA)^\dagger \\ &= B^\dagger A^\dagger - A^\dagger B^\dagger \\ &= BA - AB \\ &= -[A, B]. \end{aligned}$$

Therefore,

$$\boxed{[A, B]^\dagger = -[A, B]},$$

so the commutator is anti-Hermitian. Its expectation value is consequently purely imaginary:

$$\langle [A, B] \rangle^* = \langle [A, B]^\dagger \rangle = -\langle [A, B] \rangle.$$

(b) Application of the Schwarz inequality.

Define

$$|\chi_1\rangle = \bar{A}|\psi\rangle, \quad |\chi_2\rangle = \bar{B}|\psi\rangle.$$

Then

$$\begin{aligned} \langle \chi_1 | \chi_1 \rangle &= \langle \psi | \bar{A}^\dagger \bar{A} | \psi \rangle = \langle \bar{A}^2 \rangle = (\Delta A)^2, \\ \langle \chi_2 | \chi_2 \rangle &= \langle \psi | \bar{B}^\dagger \bar{B} | \psi \rangle = \langle \bar{B}^2 \rangle = (\Delta B)^2. \end{aligned}$$

The Schwarz inequality therefore gives

$$\begin{aligned}(\Delta A)^2(\Delta B)^2 &= \langle \chi_1 | \chi_1 \rangle \langle \chi_2 | \chi_2 \rangle \\ &\geq |\langle \chi_1 | \chi_2 \rangle|^2 \\ &= \left| \langle \psi | \bar{A} \bar{B} | \psi \rangle \right|^2.\end{aligned}$$

Hence,

$$\boxed{(\Delta A)^2(\Delta B)^2 \geq \left| \langle \bar{A} \bar{B} \rangle \right|^2.}$$

(c) Decomposition into commutator and anticommutator.

Apply the identity

$$2\bar{A}\bar{B} = \{\bar{A}, \bar{B}\} + [\bar{A}, \bar{B}],$$

where

$$\{\bar{A}, \bar{B}\} = \bar{A}\bar{B} + \bar{B}\bar{A}.$$

Since multiples of the identity commute with every operator,

$$[\bar{A}, \bar{B}] = [A, B].$$

Therefore,

$$\langle \bar{A} \bar{B} \rangle = \frac{1}{2} \langle \{\bar{A}, \bar{B}\} \rangle + \frac{1}{2} \langle [A, B] \rangle.$$

The anticommutator is Hermitian because

$$\{\bar{A}, \bar{B}\}^\dagger = \bar{B}\bar{A} + \bar{A}\bar{B} = \{\bar{A}, \bar{B}\}.$$

Thus,

$$\langle \{\bar{A}, \bar{B}\} \rangle \in \mathbb{R}.$$

By part (a),

$$\langle [A, B] \rangle$$

is purely imaginary. The two terms are therefore respectively real and purely imaginary, so their cross terms vanish when the modulus is squared:

$$\left| \langle \bar{A} \bar{B} \rangle \right|^2 = \frac{1}{4} \left| \langle \{\bar{A}, \bar{B}\} \rangle \right|^2 + \frac{1}{4} |\langle [A, B] \rangle|^2.$$

(d) Positivity of the anticommutator term.

Substituting this result into the Schwarz inequality gives the stronger bound

$$(\Delta A)^2(\Delta B)^2 \geq \frac{1}{4} \left| \langle \{\bar{A}, \bar{B}\} \rangle \right|^2 + \frac{1}{4} |\langle [A, B] \rangle|^2.$$

The term containing the anticommutator is an absolute square and is therefore nonnegative:

$$\frac{1}{4} \left| \langle \{\bar{A}, \bar{B}\} \rangle \right|^2 \geq 0.$$

Dropping this nonnegative term produces a weaker but still valid inequality:

$$\boxed{(\Delta A)^2(\Delta B)^2 \geq \frac{1}{4} |\langle [A, B] \rangle|^2.}$$

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3. Show that if the Hamiltonian commutes with the parity operator, the dipole moments of nondegenerate states are zero. The dipole moment operator is defined as

$$\mathbf{d} = \sum_{i=1}^N q_i \mathbf{R}_i,$$

where q_i is the charge, $\mathbf{R}_i$ is the position operator of the i th particle, and N is the number of particles.

Solution.

Let $\mathcal{P}$ denote the parity operator, and let

$$H|n\rangle = E_n|n\rangle$$

be a nondegenerate energy eigenstate. Since

$$[H, \mathcal{P}] = 0,$$

we have

$$\begin{aligned} H(\mathcal{P}|n\rangle) &= \mathcal{P}H|n\rangle \\ &= E_n\mathcal{P}|n\rangle. \end{aligned}$$

Thus, $\mathcal{P}|n\rangle$ is also an eigenstate of H with energy E_n . Because this energy is nondegenerate, $\mathcal{P}|n\rangle$ must be proportional to $|n\rangle$:

$$\mathcal{P}|n\rangle = \eta_n|n\rangle.$$

The parity operator satisfies

$$\mathcal{P}^\dagger = \mathcal{P}, \quad \mathcal{P}^2 = I,$$

so its eigenvalues are

$$\eta_n = \pm 1.$$

Therefore, every nondegenerate energy eigenstate is also a parity eigenstate.

Under parity, each position operator changes sign:

$$\mathcal{P}^\dagger \mathbf{R}_i \mathcal{P} = -\mathbf{R}_i.$$

The charges q_i are unchanged by parity, so the dipole operator transforms according to

$$\begin{aligned}\mathcal{P}^\dagger \mathbf{d} \mathcal{P} &= \mathcal{P}^\dagger \left(\sum_{i=1}^N q_i \mathbf{R}_i \right) \mathcal{P} \\ &= \sum_{i=1}^N q_i \mathcal{P}^\dagger \mathbf{R}_i \mathcal{P} \\ &= - \sum_{i=1}^N q_i \mathbf{R}_i \\ &= -\mathbf{d}.\end{aligned}$$

Thus, the dipole operator is odd under parity.

The dipole moment of the state $|n\rangle$ is the expectation value

$$\langle \mathbf{d} \rangle_n = \langle n | \mathbf{d} | n \rangle.$$

Now consider

$$\langle n | \mathcal{P}^\dagger \mathbf{d} \mathcal{P} | n \rangle.$$

Using the fact that $|n\rangle$ is a parity eigenstate gives

$$\begin{aligned}\langle n | \mathcal{P}^\dagger \mathbf{d} \mathcal{P} | n \rangle &= \eta_n^* \eta_n \langle n | \mathbf{d} | n \rangle \\ &= \langle \mathbf{d} \rangle_n.\end{aligned}$$

On the other hand, since the dipole operator is odd under parity,

$$\begin{aligned}\langle n | \mathcal{P}^\dagger \mathbf{d} \mathcal{P} | n \rangle &= -\langle n | \mathbf{d} | n \rangle \\ &= -\langle \mathbf{d} \rangle_n.\end{aligned}$$

Consequently,

$$\langle \mathbf{d} \rangle_n = -\langle \mathbf{d} \rangle_n,$$

which implies

$$\boxed{\langle n | \mathbf{d} | n \rangle = 0.}$$

The dipole operator itself is not zero. Rather, its expectation value vanishes in every nondegenerate energy eigenstate of a parity-invariant Hamiltonian.

4. In the Raman (absorption) rotational spectrum of $^{35}\text{Cl}_2$, the consecutive lines have alternating intensities with a ratio 3:5. Assume that $^{35}\text{Cl}_2$ is a rigid rotor (two nuclei at a fixed separation), the observed rotational transitions correspond to energy levels with large rigid-rotor angular quantum numbers J , and that temperature is so high that all states corresponding to these levels are nearly uniformly populated. Raman selection rule is $J \rightarrow J + 2$. What is the spin of the ^{35}Cl nucleus? Hints:

$$E_J = BJ(J+1), \quad Y_J^M(\pi - \theta, \pi + \phi) = (-1)^J Y_J^M(\theta, \phi),$$

where B is a constant.

- (a) Explain the shape of the spectrum and illustrate it on a figure. Then explain the relation between line intensity and degeneracy of energy levels.
- (b) Determine the degeneracy of energy levels with even and odd angular quantum numbers (you have to consider both fermionic and bosonic nuclei, since

Solution.

(a) Shape and intensity of the spectrum.

The energy absorbed in a transition $J \rightarrow J + 2$ is

$$\begin{aligned}\Delta E_J &= E_{J+2} - E_J \\ &= B(J+2)(J+3) - BJ(J+1) \\ &= B[(J+2)(J+3) - J(J+1)] \\ &= B(4J+6).\end{aligned}$$

Thus,

$$\boxed{\Delta E_J = B(4J+6).}$$

For transitions originating from consecutive values J and $J+1$,

$$\begin{aligned}\Delta E_{J+1} - \Delta E_J &= B[4(J+1)+6] - B(4J+6) \\ &= 4B.\end{aligned}$$

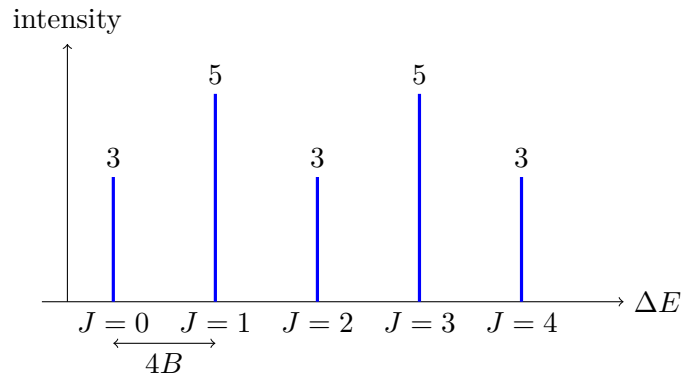
Therefore, the Raman lines are equally spaced, with separation

$$\boxed{4B.}$$

The first few line positions are

$$\frac{\Delta E_J}{B} = 6, 10, 14, 18, 22, \dots$$

Because the selection rule $J \rightarrow J + 2$ preserves the parity of J , the lines originating from even and odd J values form two interlaced families.



Suppose the level with rotational quantum number J has total degeneracy G_J . The population of each individual state in that level is proportional to $e^{-\beta E_J}$. Therefore, the total population of the level is

$$N_J \propto G_J e^{-\beta E_J}.$$

At the stated high temperature, the Boltzmann factors for neighboring levels are approximately equal. The intensity of an absorption line is proportional to the population of its initial level multiplied by the transition probability. For consecutive transitions at large J , the transition probabilities vary only slowly and may be treated as approximately equal. Consequently,

$$\boxed{\mathcal{I}_{J \rightarrow J+2} \propto G_J.}$$

A more highly degenerate level contains more approximately equally populated initial states and therefore produces a stronger absorption line. The alternating intensities thus indicate different degeneracies for even and odd values of J .

(b) Degeneracies and the nuclear spin.

Let I be the spin of one ^{35}Cl nucleus. The molecular state may be written as

$$|\Psi\rangle = |\psi_{\text{el,vib}}\rangle \otimes |JM\rangle_{\text{rot}} \otimes |\chi\rangle_{\text{ns}},$$

where $|\chi\rangle_{\text{ns}}$ is the two-nucleus spin state. The electronic and vibrational factor is taken to be symmetric under interchange of the two nuclei.

Define the internuclear vector

$$\mathbf{r} = \mathbf{R}_1 - \mathbf{R}_2.$$

Interchanging the two identical nuclei gives

$$\mathbf{r} \longrightarrow -\mathbf{r},$$

which corresponds to

$$(\theta, \phi) \longrightarrow (\pi - \theta, \pi + \phi).$$

Using the supplied identity,

$$Y_J^M(\pi - \theta, \pi + \phi) = (-1)^J Y_J^M(\theta, \phi),$$

we obtain

$$P_{12}|JM\rangle_{\text{rot}} = (-1)^J |JM\rangle_{\text{rot}}.$$

Therefore,

$$\boxed{\begin{array}{ll} J \text{ even :} & |JM\rangle_{\text{rot}} \text{ is symmetric,} \\ J \text{ odd :} & |JM\rangle_{\text{rot}} \text{ is antisymmetric.} \end{array}}$$

A single nucleus of spin I has

$$n = 2I + 1$$

possible spin states. The two-nucleus spin space therefore has dimension

$$n^2 = (2I + 1)^2.$$

For $m_1 \neq m_2$, the symmetric and antisymmetric combinations are

$$|m_1, m_2\rangle_{\pm} = \frac{1}{\sqrt{2}} (|I, m_1\rangle \otimes |I, m_2\rangle \pm |I, m_2\rangle \otimes |I, m_1\rangle).$$

There are

$$\binom{n}{2} = \frac{n(n-1)}{2}$$

unordered pairs with $m_1 \neq m_2$, giving the same number of symmetric and antisymmetric combinations. There are also n symmetric states of the form

$$|I, m\rangle \otimes |I, m\rangle,$$

but no corresponding antisymmetric states.

Thus, the number of symmetric nuclear-spin states is

$$\begin{aligned} g_s &= n + \frac{n(n-1)}{2} \\ &= \frac{n(n+1)}{2} \\ &= (2I+1)(I+1), \end{aligned}$$

while the number of antisymmetric nuclear-spin states is

$$\begin{aligned} g_a &= \frac{n(n-1)}{2} \\ &= (2I+1)I. \end{aligned}$$

As a check,

$$g_s + g_a = (2I+1)^2.$$

Each rotational level also has the magnetic degeneracy $2J+1$. For bosonic nuclei, the total molecular state must be symmetric. Therefore,

$$G_J^{(\text{B})} = \begin{cases} (2J+1)(2I+1)(I+1), & J \text{ even,} \\ (2J+1)(2I+1)I, & J \text{ odd.} \end{cases}$$

For fermionic nuclei, the total molecular state must be antisymmetric. Therefore,

$$G_J^{(\text{F})} = \begin{cases} (2J+1)(2I+1)I, & J \text{ even,} \\ (2J+1)(2I+1)(I+1), & J \text{ odd.} \end{cases}$$

For consecutive levels at large J ,

$$2J+1 \simeq 2J+3,$$

so the ordinary rotational degeneracies are approximately equal. The alternating intensity ratio is therefore determined by the nuclear-spin statistical weights:

$$\frac{\mathcal{I}_{\text{strong}}}{\mathcal{I}_{\text{weak}}} \simeq \frac{g_s}{g_a} = \frac{I+1}{I}.$$

The observed ratio is 5 : 3, so

$$\frac{I+1}{I} = \frac{5}{3}.$$

It follows that

$$3I+3 = 5I,$$

and hence

$$I = \frac{3}{2}.$$

Since $I = 3/2$ is half-integer, the ^{35}Cl nucleus is a fermion. Explicitly,

$$g_s = (2I + 1)(I + 1) = 4 \left(\frac{5}{2} \right) = 10,$$

and

$$g_a = (2I + 1)I = 4 \left(\frac{3}{2} \right) = 6.$$

For fermionic nuclei, even J must be paired with antisymmetric nuclear-spin states, while odd J must be paired with symmetric nuclear-spin states. Thus,

$$G_{J \text{ even}} = 6(2J + 1), \quad G_{J \text{ odd}} = 10(2J + 1).$$

At large J ,

$$\mathcal{I}_{\text{even}} : \mathcal{I}_{\text{odd}} \simeq 6 : 10 = 3 : 5,$$

in agreement with the observed spectrum.

we do not know a priori which case is realized) and then determine the spin.

5. The members of the family of states $|jj\rangle, \dots, |jm\rangle, \dots, |j, -j\rangle$ are often referred to as states with different orientation of angular momentum. This may lead one to believe that it is possible to rotate these states into each other, as is the case for the corresponding classical quantities. One can see that this is not so on the example of

$$|10\rangle = [0, 1, 0]^T, \quad |1, -1\rangle = [0, 0, 1]^T,$$

the eigenvectors of the matrix $\mathbb{J}_z^{(1)}$, where a naive association with $\hat{\mathbf{y}}$ and $\hat{\mathbf{z}}$, respectively, would make one to believe that one can transform one state into another by a rotation around $\hat{\mathbf{x}}$. Show that

$$\begin{aligned} |10\rangle &\neq \mathbb{D}^{(1)}[R_{\theta_x \hat{\mathbf{x}}}] |1, -1\rangle \\ &= \left[(\cos \theta_x - 1) \left(\frac{\mathbb{J}_x^{(1)}}{\hbar} \right)^2 - i \sin \theta_x \frac{\mathbb{J}_x^{(1)}}{\hbar} + \mathbb{I}^{(1)} \right] |1, -1\rangle \end{aligned}$$

for any θ_x . Do not derive the given expression for $\mathbb{D}^{(1)}[R_{\theta_x \hat{\mathbf{x}}}]$, but do find $\mathbb{J}_x^{(1)}$.

Solution.

Work in the ordered J_z basis

$$\mathcal{B} = \{|1, 1\rangle, |1, 0\rangle, |1, -1\rangle\}.$$

In this basis,

$$|1, 0\rangle = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad |1, -1\rangle = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

The ladder operators act according to

$$J_{\pm}|j, m\rangle = \hbar\sqrt{j(j+1) - m(m\pm 1)}|j, m\pm 1\rangle.$$

For $j = 1$,

$$\begin{aligned} J_+|1, -1\rangle &= \sqrt{2}\hbar|1, 0\rangle, \\ J_+|1, 0\rangle &= \sqrt{2}\hbar|1, 1\rangle, \\ J_+|1, 1\rangle &= 0, \end{aligned}$$

and similarly,

$$\begin{aligned} J_-|1, 1\rangle &= \sqrt{2}\hbar|1, 0\rangle, \\ J_-|1, 0\rangle &= \sqrt{2}\hbar|1, -1\rangle, \\ J_-|1, -1\rangle &= 0. \end{aligned}$$

Therefore,

$$\mathbb{J}_+^{(1)} = \sqrt{2}\hbar \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \quad \mathbb{J}_-^{(1)} = \sqrt{2}\hbar \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

Since

$$J_x = \frac{1}{2}(J_+ + J_-),$$

the spin-one matrix representing J_x is

$$\boxed{\mathbb{J}_x^{(1)} = \frac{\hbar}{\sqrt{2}} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}}.$$

Define

$$X = \frac{\mathbb{J}_x^{(1)}}{\hbar} = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}.$$

Its square is

$$X^2 = \frac{1}{2} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{bmatrix}.$$

Acting on $|1, -1\rangle$, we find

$$X|1, -1\rangle = \frac{1}{\sqrt{2}}|1, 0\rangle = \begin{bmatrix} 0 \\ 1/\sqrt{2} \\ 0 \end{bmatrix},$$

and

$$X^2|1, -1\rangle = \frac{1}{2}(|1, 1\rangle + |1, -1\rangle) = \begin{bmatrix} 1/2 \\ 0 \\ 1/2 \end{bmatrix}.$$

Using the supplied rotation operator,

$$\mathbb{D}^{(1)}[R_{\theta_x \hat{\mathbf{x}}}] = (\cos \theta_x - 1)X^2 - i \sin \theta_x X + \mathbb{I}^{(1)},$$

we obtain

$$\begin{aligned} \mathbb{D}^{(1)}[R_{\theta_x \hat{\mathbf{x}}}]|1, -1\rangle &= (\cos \theta_x - 1)X^2|1, -1\rangle - i \sin \theta_x X|1, -1\rangle + |1, -1\rangle \\ &= \frac{\cos \theta_x - 1}{2}|1, 1\rangle - \frac{i \sin \theta_x}{\sqrt{2}}|1, 0\rangle + \frac{1 + \cos \theta_x}{2}|1, -1\rangle. \end{aligned}$$

Thus, in the J_z basis,

$$\mathbb{D}^{(1)}[R_{\theta_x \hat{\mathbf{x}}}]|1, -1\rangle = \begin{bmatrix} \frac{\cos \theta_x - 1}{2} \\ -\frac{i \sin \theta_x}{\sqrt{2}} \\ \frac{1 + \cos \theta_x}{2} \end{bmatrix}.$$

For this vector to equal

$$|1, 0\rangle = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix},$$

its first and third components would both have to vanish. The first component vanishes only if

$$\cos \theta_x = 1,$$

whereas the third component vanishes only if

$$\cos \theta_x = -1.$$

These conditions cannot hold simultaneously. Therefore,

$$|1, 0\rangle \neq \mathbb{D}^{(1)}[R_{\theta_x \hat{\mathbf{x}}}]|1, -1\rangle$$

for every θ_x .

The conclusion also holds even if states differing only by a global phase are identified. Indeed,

$$\left| \langle 1, 0 | \mathbb{D}^{(1)}[R_{\theta_x \hat{\mathbf{x}}}] | 1, -1 \rangle \right|^2 = \frac{1}{2} \sin^2 \theta_x \leq \frac{1}{2} < 1.$$

Thus, no rotation about the x axis transforms $|1, -1\rangle$ into $|1, 0\rangle$, even up to an overall phase. The labels m specify angular momentum projections along the z axis; they should not be interpreted as ordinary Cartesian directions.

6. A beam of spin-1/2 particles moving along the y axis goes through two Stern–Gerlach apparatuses, both with lower (spin down) beams blocked. The first has its $\mathbf{B}$ field along the z axis and the second has its $\mathbf{B}$ field along the x axis.
- (a) What fraction of particles leaving the first will exit the second?
 - (b) If a third filter that transmits only spin up along the z axis is introduced, what fraction of particles leaving the first will exit the third?
 - (c) If the middle filter transmits both spins up and down the x axis (no blocking), but the last one transmits only spin down the z axis, what fraction of particles leaving the first will leave the last?

You may need

$$S_x = \frac{\hbar}{2} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad S_z = \frac{\hbar}{2} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

Solution.

Use the S_z eigenbasis

$$|+z\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |-z\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Solving the eigenvalue equation for

$$S_x = \frac{\hbar}{2} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

gives

$$|+x\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} (|+z\rangle + |-z\rangle),$$

and

$$|-x\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{1}{\sqrt{2}} (|+z\rangle - |-z\rangle).$$

After leaving the first Stern–Gerlach apparatus, every transmitted particle is in the state

$$|\psi_1\rangle = |+z\rangle.$$

(a) Fraction exiting the second apparatus.

The second apparatus transmits only the spin-up state along the x axis. The probability that a particle in $|+z\rangle$ is measured in $|+x\rangle$ is

$$\begin{aligned} P(+x|+z) &= |\langle +x|+z\rangle|^2 \\ &= \left| \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right|^2 \\ &= \left| \frac{1}{\sqrt{2}} \right|^2 \\ &= \frac{1}{2}. \end{aligned}$$

Therefore,

$$\boxed{\frac{1}{2}}$$

of the particles leaving the first apparatus exit the second.

(b) A third filter transmitting only spin up along z .

Particles that exit the second apparatus are in the state $|+x\rangle$. The probability that such a particle passes the third z -up filter is

$$\begin{aligned} P(+z|+x) &= |\langle +z|+x\rangle|^2 \\ &= \left| \frac{1}{\sqrt{2}} \right|^2 \\ &= \frac{1}{2}. \end{aligned}$$

A particle must pass both the second and third filters. Hence, the fraction relative to the particles leaving the first apparatus is

$$\begin{aligned} P(+x, +z|+z) &= P(+x|+z)P(+z|+x) \\ &= \frac{1}{2} \cdot \frac{1}{2} \\ &= \frac{1}{4}. \end{aligned}$$

Therefore,

$$\boxed{\frac{1}{4}}$$

of the particles leaving the first apparatus exit the third.

(c) Both x -spin beams transmitted.

The middle Stern–Gerlach apparatus now transmits both the x -up and x -down beams. The probabilities of the two outcomes are

$$P(+x|+z) = \frac{1}{2}, \quad P(-x|+z) = \frac{1}{2}.$$

If the two spatially separated beams are not coherently recombined, the middle apparatus performs a nonselective measurement of S_x . The state after this apparatus is therefore the statistical mixture

$$\begin{aligned} \rho_2 &= \frac{1}{2}|+x\rangle\langle +x| + \frac{1}{2}|-x\rangle\langle -x| \\ &= \frac{1}{2}I. \end{aligned}$$

The projector onto spin down along the z axis is

$$P_{-z} = |-z\rangle\langle -z|.$$

Thus, the probability of passing the final z -down filter is

$$\begin{aligned} P(-z) &= \text{Tr}(P_{-z}\rho_2) \\ &= \text{Tr}\left(|-z\rangle\langle -z|\frac{1}{2}I\right) \\ &= \frac{1}{2}. \end{aligned}$$

Equivalently, adding the probabilities from the two distinguishable x -spin paths gives

$$P(-z) = \frac{1}{2}|\langle -z|+x\rangle|^2 + \frac{1}{2}|\langle -z|-x\rangle|^2 = \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2}.$$

Therefore,

$$\boxed{\frac{1}{2}}$$

of the particles leaving the first apparatus leave the final apparatus.

This answer assumes that the middle Stern–Gerlach apparatus leaves the two x -spin beams distinguishable, as is standard for a nonselective Stern–Gerlach measurement. If the two beams were instead recombined coherently so that the middle apparatus had no net effect, the state would remain $|+z\rangle$, and the final z -down probability would be zero.

7. Denote by $\mathbf{J}_1$ and $\mathbf{J}_2$ angular momentum operators of particles 1 and 2.

- (a) Invoke the postulate on the construction of quantum operators from classical quantities to define the total angular momentum operator $\mathbf{J} = \mathbf{J}_1 + \mathbf{J}_2$ (assume for a moment that these are orbital angular momentum operators).
- (b) Show that $\mathbf{J} \times \mathbf{J} = i\hbar\mathbf{J}$ (note that $\mathbf{J}_i \times \mathbf{J}_i = i\hbar\mathbf{J}_i$ by our assumptions) and, consequently, $[J^2, J_z] = 0$.
- (c) Consider the eigenfunctions $|j_i m_i\rangle$ of J_i^2 and of J_{iz} with j_1 and j_2 fixed. Prove that two of the product functions $|j_1 m_1\rangle \otimes |j_2 m_2\rangle$ are eigenfunctions of J^2 and J_z . Specify quantum numbers j and m for these functions.
- (d) We can now produce from these functions $2(j_1 + j_2) + 1$ eigenfunctions of J^2 and of J_z by the action of the $J_{\pm}$ operators. In the remaining part of the space, find the eigenfunction of J^2 and J_z with the highest j (do not derive an explicit expression for this function).
- (e) Show that this procedure can be continued until getting eigenfunctions of J^2 with $j = |j_1 - j_2|$, but not further.

Solution.

(a) Definition of the total angular momentum.

Classically, the total orbital angular momentum of two particles is the sum of their individual orbital angular momenta:

$$\mathbf{J}_{\text{cl}} = \mathbf{r}_1 \times \mathbf{p}_1 + \mathbf{r}_2 \times \mathbf{p}_2.$$

The correspondence postulate replaces the classical quantities by their quantum operators. Thus, on the product Hilbert space $\mathcal{H}_1 \otimes \mathcal{H}_2$, the total angular momentum operator is

$$\mathbf{J} = \mathbf{J}_1 \otimes I_2 + I_1 \otimes \mathbf{J}_2.$$

In the following, the identity operators will be left implicit, so that

$$\mathbf{J} = \mathbf{J}_1 + \mathbf{J}_2.$$

Operators acting on different particles commute:

$$[J_{1a}, J_{2b}] = 0$$

for all Cartesian components a, b .

(b) Angular-momentum algebra of $\mathbf{J}$.

Expanding the operator cross product gives

$$\mathbf{J} \times \mathbf{J} = \mathbf{J}_1 \times \mathbf{J}_1 + \mathbf{J}_1 \times \mathbf{J}_2 + \mathbf{J}_2 \times \mathbf{J}_1 + \mathbf{J}_2 \times \mathbf{J}_2.$$

The mixed terms cancel. Indeed,

$$\begin{aligned} (\mathbf{J}_1 \times \mathbf{J}_2 + \mathbf{J}_2 \times \mathbf{J}_1)_k &= \sum_{\ell, m} \varepsilon_{k\ell m} (J_{1\ell} J_{2m} + J_{2\ell} J_{1m}) \\ &= \sum_{\ell, m} \varepsilon_{k\ell m} J_{1\ell} J_{2m} + \sum_{\ell, m} \varepsilon_{km\ell} J_{2m} J_{1\ell} \\ &= \sum_{\ell, m} \varepsilon_{k\ell m} J_{1\ell} J_{2m} - \sum_{\ell, m} \varepsilon_{k\ell m} J_{1\ell} J_{2m} \\ &= 0, \end{aligned}$$

where operators belonging to different particles were commuted in the third line.

Using

$$\mathbf{J}_i \times \mathbf{J}_i = i\hbar \mathbf{J}_i,$$

we therefore obtain

$$\begin{aligned} \mathbf{J} \times \mathbf{J} &= i\hbar \mathbf{J}_1 + i\hbar \mathbf{J}_2 \\ &= i\hbar (\mathbf{J}_1 + \mathbf{J}_2). \end{aligned}$$

Hence,

$$\mathbf{J} \times \mathbf{J} = i\hbar \mathbf{J}.$$

Equivalently, the components satisfy

$$[J_a, J_b] = i\hbar \sum_c \varepsilon_{abc} J_c.$$

It follows that J^2 commutes with J_z :

$$\begin{aligned} [J^2, J_z] &= [J_x^2 + J_y^2 + J_z^2, J_z] \\ &= J_x [J_x, J_z] + [J_x, J_z] J_x + J_y [J_y, J_z] + [J_y, J_z] J_y \\ &= -i\hbar (J_x J_y + J_y J_x) + i\hbar (J_y J_x + J_x J_y) \\ &= 0. \end{aligned}$$

Therefore,

$$\boxed{[J^2, J_z] = 0.}$$

(c) Product states that are eigenstates of J^2 .

For fixed j_1 and j_2 , define

$$|m_1, m_2\rangle = |j_1, m_1\rangle \otimes |j_2, m_2\rangle.$$

Since

$$J_z = J_{1z} \otimes I_2 + I_1 \otimes J_{2z},$$

every product state is an eigenstate of J_z :

$$J_z|m_1, m_2\rangle = \hbar(m_1 + m_2)|m_1, m_2\rangle.$$

The largest possible value of $m = m_1 + m_2$ is

$$m_{\max} = j_1 + j_2.$$

There is only one product state with this value:

$$|\Phi_+\rangle = |j_1, j_1\rangle \otimes |j_2, j_2\rangle.$$

The total raising operator is

$$J_+ = J_{1+} \otimes I_2 + I_1 \otimes J_{2+}.$$

Since each factor of $|\Phi_+\rangle$ is already a highest-weight state,

$$J_+|\Phi_+\rangle = 0.$$

Using

$$J^2 = J_- J_+ + J_z^2 + \hbar J_z,$$

we find

$$\begin{aligned} J^2|\Phi_+\rangle &= (J_z^2 + \hbar J_z)|\Phi_+\rangle \\ &= \hbar^2(j_1 + j_2)(j_1 + j_2 + 1)|\Phi_+\rangle. \end{aligned}$$

Thus,

$$\boxed{|\Phi_+\rangle = |j_1 + j_2, j_1 + j_2\rangle,}$$

so this state has

$$j = j_1 + j_2, \quad m = j_1 + j_2.$$

Similarly, the smallest possible value of m is

$$m_{\min} = -(j_1 + j_2),$$

and the unique product state with this value is

$$|\Phi_-\rangle = |j_1, -j_1\rangle \otimes |j_2, -j_2\rangle.$$

It satisfies

$$J_-|\Phi_-\rangle = 0.$$

Using

$$J^2 = J_+ J_- + J_z^2 - \hbar J_z,$$

we obtain

$$J^2 |\Phi_-\rangle = \hbar^2 (j_1 + j_2)(j_1 + j_2 + 1) |\Phi_-\rangle.$$

Therefore,

$$\boxed{|\Phi_-\rangle = |j_1 + j_2, -(j_1 + j_2)\rangle},$$

with

$$j = j_1 + j_2, \quad m = -(j_1 + j_2).$$

(d) Construction of the next multiplet.

Starting with the highest-weight state $|\Phi_+\rangle$, repeated action of J_- produces the states

$$|j_1 + j_2, m\rangle \propto J_-^{j_1 + j_2 - m} |\Phi_+\rangle,$$

where

$$m = j_1 + j_2, j_1 + j_2 - 1, \dots, -(j_1 + j_2).$$

Since

$$[J^2, J_-] = 0,$$

all of these states have

$$j = j_1 + j_2.$$

The first multiplet therefore contains

$$2(j_1 + j_2) + 1$$

states.

Now consider the subspace with

$$m = j_1 + j_2 - 1.$$

It is spanned by the two product states

$$|j_1, j_1 - 1\rangle \otimes |j_2, j_2\rangle$$

and

$$|j_1, j_1\rangle \otimes |j_2, j_2 - 1\rangle.$$

One linear combination is

$$J_- |\Phi_+\rangle$$

and belongs to the $j = j_1 + j_2$ multiplet. Choose a normalized state

$$|\Phi_1\rangle$$

in this two-dimensional subspace that is orthogonal to $J_- |\Phi_+\rangle$. No explicit expression for $|\Phi_1\rangle$ is required.

Because $J_+ |\Phi_1\rangle$ would lie in the one-dimensional subspace with $m = j_1 + j_2$, it must be proportional to $|\Phi_+\rangle$. However,

$$\langle \Phi_+ | J_+ | \Phi_1 \rangle = \langle J_- \Phi_+ | \Phi_1 \rangle = 0.$$

Hence,

$$J_+|\Phi_1\rangle = 0.$$

Thus, $|\Phi_1\rangle$ is a new highest-weight state with

$$m = j_1 + j_2 - 1.$$

Using

$$J^2 = J_- J_+ + J_z^2 + \hbar J_z,$$

we find

$$J^2|\Phi_1\rangle = \hbar^2(j_1 + j_2 - 1)(j_1 + j_2)|\Phi_1\rangle.$$

Therefore,

$$\boxed{j = j_1 + j_2 - 1, \quad m = j_1 + j_2 - 1.}$$

Repeated application of J_- to this state generates the complete $j = j_1 + j_2 - 1$ multiplet.

(e) Continuation and termination of the procedure.

Assume without loss of generality that

$$j_1 \geq j_2.$$

At the r th stage, consider the product-state subspace with

$$m = j_1 + j_2 - r.$$

For

$$0 \leq r \leq 2j_2,$$

there are $r + 1$ product states with this value of m . They correspond to all nonnegative pairs r_1, r_2 satisfying

$$r_1 + r_2 = r, \quad m_1 = j_1 - r_1, \quad m_2 = j_2 - r_2.$$

The r previously constructed multiplets contribute r states to this subspace. Consequently, one orthogonal state remains. As in part (d), this remaining state is annihilated by J_+ and is therefore a new highest-weight state with

$$j = m = j_1 + j_2 - r.$$

The construction consequently produces

$$j = j_1 + j_2, j_1 + j_2 - 1, j_1 + j_2 - 2, \dots$$

The final new multiplet occurs when $r = 2j_2$, for which

$$j = j_1 + j_2 - 2j_2 = j_1 - j_2.$$

Restoring symmetry between j_1 and j_2 , the smallest allowed total angular momentum is

$$j_{\min} = |j_1 - j_2|.$$

The dimensions provide a final check:

$$\sum_{j=|j_1-j_2|}^{j_1+j_2} (2j+1) = (2j_1+1)(2j_2+1),$$

which is exactly the dimension of the product space

$$\dim(\mathcal{H}_1 \otimes \mathcal{H}_2) = (2j_1 + 1)(2j_2 + 1).$$

All product-space states have therefore been exhausted, so the procedure cannot continue to any smaller j .

Hence, the possible total angular-momentum quantum numbers are

$$j = j_1 + j_2, j_1 + j_2 - 1, \dots, |j_1 - j_2|.$$

For each value of j ,

$$m = -j, -j + 1, \dots, j.$$

8. By definition, $\mathbf{V}$ is called a vector operator if its rotated components

$$\tilde{V}_i = U^\dagger[R] V_i U[R],$$

with $U[R]$ being the operator corresponding to the rotation R , can be written in terms of the original components in the following way:

$$\tilde{V}_i = \sum_{j=1}^3 R_{ij} V_j, \tag{1}$$

where R_{ij} are the elements of the rotation matrix (acting on $[x, y, z]^T$).

- (viii) For an infinitesimal rotation $\epsilon \hat{\boldsymbol{\theta}}$, i.e., a rotation by an angle ϵ about axis $\hat{\boldsymbol{\theta}}$, show that

$$\tilde{V}_i = V_i + (\epsilon \hat{\boldsymbol{\theta}} \times \mathbf{V})_i.$$

- (viii) Feed in

$$U[R] = 1 - \frac{i}{\hbar} \epsilon \hat{\boldsymbol{\theta}} \cdot \mathbf{L}$$

into Eq. (1) and deduce that

$$[V_i, L_j] = i\hbar \sum_k \epsilon_{ijk} V_k.$$

(This gives an alternative derivation of the commutation relations between operators L_i or between P_i and L_i .)

Solution.

Let the infinitesimal rotation have angle ϵ about the unit vector $\hat{\boldsymbol{\theta}}$.

(h) For a finite rotation, Rodrigues' formula gives

$$R\mathbf{v} = \mathbf{v} \cos \epsilon + (\hat{\boldsymbol{\theta}} \times \mathbf{v}) \sin \epsilon + \hat{\boldsymbol{\theta}}(\hat{\boldsymbol{\theta}} \cdot \mathbf{v})(1 - \cos \epsilon).$$

Using

$$\cos \epsilon = 1 + O(\epsilon^2), \quad \sin \epsilon = \epsilon + O(\epsilon^3),$$

we obtain

$$R\mathbf{v} = \mathbf{v} + \epsilon \hat{\boldsymbol{\theta}} \times \mathbf{v} + O(\epsilon^2).$$

Equivalently, the infinitesimal rotation matrix is

$$R_{ij} = \delta_{ij} - \epsilon \sum_k \varepsilon_{ijk} \hat{\theta}_k + O(\epsilon^2).$$

Since $\mathbf{V}$ is a vector operator,

$$\tilde{V}_i = \sum_j R_{ij} V_j.$$

Therefore,

$$\begin{aligned} \tilde{V}_i &= V_i - \epsilon \sum_{j,k} \varepsilon_{ijk} \hat{\theta}_k V_j + O(\epsilon^2) \\ &= V_i + \epsilon \sum_{j,k} \varepsilon_{ijk} \hat{\theta}_j V_k + O(\epsilon^2), \end{aligned}$$

where the dummy indices j and k were interchanged in the second line and the antisymmetry of ε_{ijk} was used. Thus,

$$\boxed{\tilde{V}_i = V_i + (\epsilon \hat{\boldsymbol{\theta}} \times \mathbf{V})_i + O(\epsilon^2)}.$$

(h) The rotation operator for an infinitesimal rotation is

$$U[R] = \mathbf{1} - \frac{i\epsilon}{\hbar} \hat{\boldsymbol{\theta}} \cdot \mathbf{L} + O(\epsilon^2),$$

and hence

$$U^\dagger[R] = \mathbf{1} + \frac{i\epsilon}{\hbar} \hat{\boldsymbol{\theta}} \cdot \mathbf{L} + O(\epsilon^2).$$

Writing

$$\hat{\boldsymbol{\theta}} \cdot \mathbf{L} = \sum_j \hat{\theta}_j L_j,$$

the transformed component is

$$\begin{aligned} \tilde{V}_i &= U^\dagger[R] V_i U[R] \\ &= \left(\mathbf{1} + \frac{i\epsilon}{\hbar} \sum_j \hat{\theta}_j L_j \right) V_i \left(\mathbf{1} - \frac{i\epsilon}{\hbar} \sum_k \hat{\theta}_k L_k \right) + O(\epsilon^2) \\ &= V_i + \frac{i\epsilon}{\hbar} \sum_j \hat{\theta}_j (L_j V_i - V_i L_j) + O(\epsilon^2) \\ &= V_i + \frac{i\epsilon}{\hbar} \sum_j \hat{\theta}_j [L_j, V_i] + O(\epsilon^2). \end{aligned}$$

From part (a), the same transformation must satisfy

$$\tilde{V}_i = V_i + \epsilon \sum_{j,k} \varepsilon_{ijk} \hat{\theta}_j V_k + O(\epsilon^2).$$

Comparing the coefficients of the arbitrary components $\hat{\theta}_j$ gives

$$\frac{i}{\hbar} [L_j, V_i] = \sum_k \varepsilon_{ijk} V_k.$$

Therefore,

$$[L_j, V_i] = -i\hbar \sum_k \varepsilon_{ijk} V_k,$$

or, reversing the order of the commutator,

$$\boxed{[V_i, L_j] = i\hbar \sum_k \varepsilon_{ijk} V_k.}$$

As checks, choosing $\mathbf{V} = \mathbf{L}$ gives

$$\boxed{[L_i, L_j] = i\hbar \sum_k \varepsilon_{ijk} L_k,}$$

while choosing $\mathbf{V} = \mathbf{P}$ gives

$$\boxed{[P_i, L_j] = i\hbar \sum_k \varepsilon_{ijk} P_k.}$$

Thus the familiar angular-momentum commutation relations follow directly from the fact that $\mathbf{L}$ and $\mathbf{P}$ transform as vector operators.

9. Consider a particle in a central potential. We know that the eigenstates of the Hamiltonian can be written in the form

$$\Psi_{\alpha,l,m}(\mathbf{r}) = \frac{1}{r} U_{\alpha,l}(r) Y_l^m(\theta, \phi),$$

where Y_l^m are spherical harmonics, and correspond to the eigenenergies $E_{\alpha,l}$ (do not show this). Use the variational principle to show that

$$E_{\alpha_{\min}(l),l} < E_{\alpha_{\min}(l+1),l+1},$$

where $E_{\alpha_{\min}(l),l}$ is the lowest energy eigenvalue with a given l . Assume that $\Psi_{\alpha,l,m}(\mathbf{r})$ are normalized. You may use without proof the formula

$$\nabla^2 = \frac{1}{r^2} \frac{\partial}{\partial r} r^2 \frac{\partial}{\partial r} - \frac{L^2}{\hbar^2 r^2},$$

where $\mathbf{L}$ is the angular momentum operator.

Solution.

Denote the mass of the particle by M , so that it is not confused with the magnetic quantum number m . The Hamiltonian is

$$H = -\frac{\hbar^2}{2M}\nabla^2 + V(r).$$

For

$$\Psi_{\alpha,l,m}(\mathbf{r}) = \frac{1}{r}U_{\alpha,l}(r)Y_l^m(\theta, \phi),$$

we use

$$L^2 Y_l^m = \hbar^2 l(l+1) Y_l^m$$

and

$$\frac{1}{r^2} \frac{d}{dr} \left[r^2 \frac{d}{dr} \left(\frac{U}{r} \right) \right] = \frac{1}{r} \frac{d^2 U}{dr^2}.$$

It follows that

$$H\Psi_{\alpha,l,m} = \frac{1}{r} \left[-\frac{\hbar^2}{2M} \frac{d^2}{dr^2} + V(r) + \frac{\hbar^2 l(l+1)}{2Mr^2} \right] U_{\alpha,l}(r) Y_l^m.$$

Thus, in the subspace of fixed l , the reduced radial Hamiltonian is

$$H_l = -\frac{\hbar^2}{2M} \frac{d^2}{dr^2} + V(r) + \frac{\hbar^2 l(l+1)}{2Mr^2}.$$

Since the spherical harmonics are normalized,

$$\int d\Omega |Y_l^m(\theta, \phi)|^2 = 1,$$

normalization of the full wavefunction implies

$$1 = \int d^3r |\Psi_{\alpha,l,m}(\mathbf{r})|^2 = \int_0^\infty dr |U_{\alpha,l}(r)|^2.$$

Define the lowest energy in the fixed- l subspace by

$$E_l^{\min} \equiv \min_{\alpha} E_{\alpha,l} = E_{\alpha_{\min}(l),l},$$

and let

$$U_{l+1}(r) \equiv U_{\alpha_{\min}(l+1),l+1}(r)$$

be the normalized reduced radial wavefunction corresponding to the lowest-energy state with angular momentum $l+1$. It satisfies

$$H_{l+1} U_{l+1} = E_{l+1}^{\min} U_{l+1}.$$

We now use the same radial function to construct a normalized trial state in the angular-momentum- l subspace:

$$\Phi_{l,m}(\mathbf{r}) = \frac{1}{r} U_{l+1}(r) Y_l^m(\theta, \phi).$$

By the variational principle restricted to the fixed- l subspace,

$$E_l^{\min} \leq \langle \Phi_{l,m} | H | \Phi_{l,m} \rangle = \langle U_{l+1} | H_l | U_{l+1} \rangle.$$

The two radial Hamiltonians differ only in their centrifugal terms:

$$\begin{aligned} H_{l+1} - H_l &= \frac{\hbar^2}{2Mr^2} [(l+1)(l+2) - l(l+1)] \\ &= \frac{\hbar^2(l+1)}{Mr^2}. \end{aligned}$$

Therefore,

$$H_l = H_{l+1} - \frac{\hbar^2(l+1)}{Mr^2}.$$

Using U_{l+1} as the trial radial wavefunction gives

$$\begin{aligned} \langle U_{l+1} | H_l | U_{l+1} \rangle &= \langle U_{l+1} | H_{l+1} | U_{l+1} \rangle - \frac{\hbar^2(l+1)}{M} \int_0^\infty \frac{|U_{l+1}(r)|^2}{r^2} dr \\ &= E_{l+1}^{\min} - \frac{\hbar^2(l+1)}{M} \int_0^\infty \frac{|U_{l+1}(r)|^2}{r^2} dr. \end{aligned}$$

Because U_{l+1} is normalized and not identically zero,

$$\int_0^\infty \frac{|U_{l+1}(r)|^2}{r^2} dr > 0.$$

Consequently,

$$\langle U_{l+1} | H_l | U_{l+1} \rangle < E_{l+1}^{\min}.$$

Combining this strict inequality with the variational bound yields

$$E_l^{\min} \leq \langle U_{l+1} | H_l | U_{l+1} \rangle < E_{l+1}^{\min}.$$

Hence,

$$\boxed{E_{\alpha_{\min}(l),l} < E_{\alpha_{\min}(l+1),l+1}}.$$

Thus the lowest energy available at fixed orbital angular momentum increases strictly as l increases.

10. Consider the hydrogen atom with energy equal to the first excited state energy (quantum number $n = 2$) in a uniform electric field along the z axis (Stark effect).
 - (a) Show that the standard second-order perturbation expression has an undetermined value due to a division by zero.
 - (b) Propose a way to avoid this difficulty and show that it leads to a nonzero contribution already in the first-order of perturbation theory.
 - (c) Determine the zeroth-order wave vectors.

- (d) Plot the behavior of the $n = 2$ states as functions of the strength of the electric field and mark the linear and quadratic Stark effect regions. Do not derive the complete expressions for the first-order or second-order energy corrections.

Solution.

Let the external electric field be

$$\mathbf{E}_{\text{ext}} = \mathcal{E} \hat{\mathbf{z}}.$$

For an electron of charge $-e$, where $e > 0$, the perturbation is

$$H' = e\mathcal{E}z.$$

We neglect spin, since H' acts only on the spatial wavefunction. The four degenerate $n = 2$ states are

$$|200\rangle, \quad |210\rangle, \quad |211\rangle, \quad |21, -1\rangle,$$

all with unperturbed energy $E_2^{(0)}$.

- (j) Ordinary nondegenerate perturbation theory gives

$$E_a^{(2)} = \sum_{b \neq a} \frac{|\langle b | H' | a \rangle|^2}{E_a^{(0)} - E_b^{(0)}}.$$

Every diagonal first-order correction vanishes because z is odd under parity:

$$\langle nlm | H' | nlm \rangle = e\mathcal{E} \langle nlm | z | nlm \rangle = 0.$$

However, the off-diagonal matrix element between $|200\rangle$ and $|210\rangle$ is nonzero. Using

$$z = r \cos \theta,$$

together with

$$R_{20}(r) = \frac{1}{2\sqrt{2} a_0^{3/2}} \left(2 - \frac{r}{a_0} \right) e^{-r/(2a_0)}$$

and

$$R_{21}(r) = \frac{1}{2\sqrt{6} a_0^{3/2}} \frac{r}{a_0} e^{-r/(2a_0)},$$

we find

$$\int d\Omega Y_{00}^* \cos \theta Y_{10} = \frac{1}{\sqrt{3}},$$

and

$$\begin{aligned} \int_0^\infty R_{20}(r) R_{21}(r) r^3 dr &= \frac{a_0}{8\sqrt{3}} \int_0^\infty x^4 (2 - x) e^{-x} dx \\ &= \frac{a_0}{8\sqrt{3}} (2 \cdot 4! - 5!) \\ &= -3\sqrt{3} a_0. \end{aligned}$$

Therefore,

$$\langle 200|z|210\rangle = -3a_0$$

and hence

$$\langle 200|H'|210\rangle = -3ea_0\mathcal{E}.$$

If the nondegenerate second-order formula is applied to $|200\rangle$, it contains the term

$$E_{200}^{(2)} \supset \frac{|\langle 210|H'|200\rangle|^2}{E_2^{(0)} - E_2^{(0)}} = \frac{9e^2a_0^2\mathcal{E}^2}{0}.$$

Thus,

$$\text{nondegenerate perturbation theory produces a division by zero.}$$

The failure occurs because the perturbation couples two states that are degenerate under H_0 .

(j) We must diagonalize H' within the complete $n = 2$ degenerate subspace. Since

$$z = r\sqrt{\frac{4\pi}{3}}Y_1^0,$$

its selection rules are

$$\Delta l = \pm 1, \quad \Delta m = 0.$$

Consequently, within the $n = 2$ subspace, only $|200\rangle$ and $|210\rangle$ are coupled.

In the ordered basis

$$\{|200\rangle, |210\rangle, |211\rangle, |21, -1\rangle\},$$

the perturbation matrix is

$$W = \begin{pmatrix} \langle 200|H'|200\rangle & \langle 200|H'|210\rangle & \langle 200|H'|211\rangle & \langle 200|H'|21, -1\rangle \\ \langle 210|H'|200\rangle & \langle 210|H'|210\rangle & \langle 210|H'|211\rangle & \langle 210|H'|21, -1\rangle \\ \langle 211|H'|200\rangle & \langle 211|H'|210\rangle & \langle 211|H'|211\rangle & \langle 211|H'|21, -1\rangle \\ \langle 21, -1|H'|200\rangle & \langle 21, -1|H'|210\rangle & \langle 21, -1|H'|211\rangle & \langle 21, -1|H'|21, -1\rangle \end{pmatrix},$$

which reduces to

$$W = e\mathcal{E} \begin{pmatrix} 0 & -3a_0 & 0 & 0 \\ -3a_0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

The secular equation is

$$\det(W - \lambda\mathbf{1}) = 0,$$

or

$$\lambda^2 [\lambda^2 - (3ea_0\mathcal{E})^2] = 0.$$

Therefore, the first-order energy corrections are

$$E_+^{(1)} = +3ea_0\mathcal{E}, \quad E_-^{(1)} = -3ea_0\mathcal{E}, \quad E_{m=\pm 1}^{(1)} = 0.$$

The degeneracy is therefore partially lifted at first order. This is the linear Stark effect.

- (j) The correct zeroth-order states are the normalized eigenvectors of W , rather than the original $|200\rangle$ and $|210\rangle$ separately.

For the positive first-order correction,

$$|\psi_+^{(0)}\rangle = \frac{1}{\sqrt{2}}(|200\rangle - |210\rangle), \quad E_+^{(1)} = +3ea_0\mathcal{E}.$$

For the negative first-order correction,

$$|\psi_-^{(0)}\rangle = \frac{1}{\sqrt{2}}(|200\rangle + |210\rangle), \quad E_-^{(1)} = -3ea_0\mathcal{E}.$$

The remaining two zeroth-order states may be chosen as

$$|\psi_{+1}^{(0)}\rangle = |211\rangle, \quad |\psi_{-1}^{(0)}\rangle = |21, -1\rangle,$$

and both have zero first-order correction.

The relative signs in the two mixed states depend on the phase convention chosen for $|210\rangle$; the energy shifts do not.

- (j) For sufficiently weak fields, the two mixed states have energies

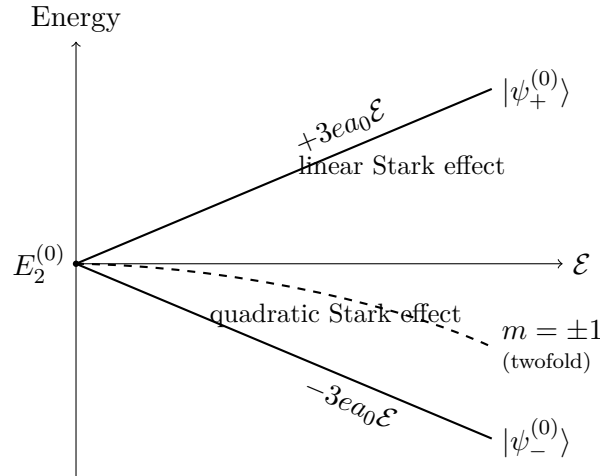
$$E_{\pm} = E_2^{(0)} \pm 3ea_0\mathcal{E} + O(\mathcal{E}^2).$$

Their leading shifts are proportional to $\mathcal{E}$, so they exhibit the linear Stark effect.

For the $m = \pm 1$ states, the first-order shift vanishes. Their leading correction is therefore second order:

$$E_{m=\pm 1} = E_2^{(0)} - C\mathcal{E}^2 + \dots, \quad C > 0.$$

These two states remain degenerate and exhibit the quadratic Stark effect.



Thus the $n = 2$ level separates into two linearly shifted branches and one twofold-degenerate branch whose leading shift is quadratic in the applied electric field.